

Parametric Computational Design and CFD Analysis of Kaplan Turbine Blade Profiles Using NACA Airfoils

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Abstract—Kaplan turbines are widely used in low-head hydropower applications due to their high efficiency and adaptability. This study presents a computational framework for blade design using NACA airfoils, followed by CFD-based comparative analysis of turbines with and without blade twist. The methodology integrates theoretical calculations, CAD modeling, and numerical simulation. The results demonstrate that blade twist significantly improves velocity distribution, pressure drop, and overall efficiency.

Index Terms—Kaplan Turbine, CFD Analysis, Blade Twist, Hydropower, Velocity Distribution, Pressure Analysis

I. INTRODUCTION

Hydropower remains one of the most reliable and sustainable sources of renewable energy due to its ability to provide continuous power generation with relatively low environmental impact. Among the various types of hydraulic turbines, Kaplan turbines are widely used for low-head and high-flow conditions, making them particularly suitable for river-based and run-of-the-river power plants. Their ability to maintain high efficiency under varying operating conditions makes them a preferred choice in modern hydropower systems.

The performance of a Kaplan turbine is strongly influenced by the design of its runner blades. These blades act as hydrofoils, converting the kinetic and pressure energy of water into mechanical rotation. Any mismatch between the incoming flow direction and blade orientation can result in energy losses, flow separation, and reduced efficiency. Therefore, careful design of blade geometry is essential to achieve optimal performance.

One of the most important aspects of blade design is the incorporation of blade twist. Since the velocity and flow conditions vary from the hub to the tip of the blade, a uniform blade angle cannot provide optimal performance throughout the span. Introducing twist allows the blade to align more effectively with local flow conditions, thereby improving energy transfer and reducing losses.

In this study, a computational approach is adopted to design Kaplan turbine blades using NACA airfoil profiles, followed by a detailed CFD analysis. Two configurations—one with blade twist and one without—are analyzed under identical conditions to understand the influence of blade geometry. The study aims to provide clear insights into how blade twist affects flow behavior, pressure distribution, and overall turbine efficiency.

II. LITERATURE REVIEW

Several studies have highlighted the importance of blade geometry in improving Kaplan turbine performance. Researchers have shown that optimized blade profiles significantly enhance efficiency and reduce hydraulic losses.

Abeykoon (2022) demonstrated that CFD-based optimization of turbine blades can lead to substantial improvements in energy conversion. The study emphasized the role of numerical simulations in predicting turbine behavior before physical implementation.

Sondhiya et al. (2018) investigated the impact of blade design on turbine efficiency and reported an improvement of over 5% using optimized geometries. Their work highlights the importance of accurate modeling in turbine design.

Recent advancements in CFD have enabled detailed analysis of flow patterns within turbines. These studies indicate that blade twist plays a critical role in aligning flow, reducing turbulence, and improving performance.

III. THEORETICAL BACKGROUND

The power output of a turbine is given by:

$$P = \rho g Q H \eta \quad (1)$$

Velocity triangles are used to determine blade angles:

$$\tan \beta_1 = \frac{V_f}{u - V_w} \quad (2)$$

$$\tan \beta_2 = \frac{V_f}{u} \quad (3)$$

The velocity triangle explains the interaction between flow velocity and blade motion.

IV. METHODOLOGY

The methodology adopted in this study follows a systematic approach that integrates theoretical design, geometric modeling, and computational fluid dynamics (CFD) analysis. The process begins with the selection of design inputs such as power output, available head, and overall efficiency. These parameters are used to calculate the discharge, flow velocity, and other essential hydraulic quantities required for turbine design.

A. Design Input Parameters

TABLE I
DESIGN INPUT PARAMETERS

Parameter	Value
Power Output	100 kW
Head	50 m
Efficiency	0.9
Number of Blades	6
Fluid Density	1000 kg/m ³
Gravity	9.81 m/s ²

Based on these calculations, two different turbine configurations are developed: one with untwisted blades and another with twisted blades. The primary objective of this comparison is to isolate the effect of blade twist on turbine performance. Both models are designed using identical input parameters to ensure that any variation in results is solely due to changes in blade geometry.

The blade geometry is created using CAD software, where careful attention is given to maintaining dimensional consistency across both models. The twisted blade configuration incorporates variation in blade angle along the radial direction, allowing better alignment with local flow conditions. Once the geometries are completed, they are exported in a format suitable for CFD analysis.

The computational analysis is then carried out using ANSYS Fluent. The domain is discretized using a tetrahedral mesh to capture complex flow behavior within the turbine. Appropriate boundary conditions are applied at the inlet and outlet, and a pressure-based solver with the k-epsilon turbulence model is used to simulate turbulent flow conditions. This structured methodology ensures accurate and reliable comparison between the two turbine designs.

V. CAD MODELING

Two turbine configurations were created.

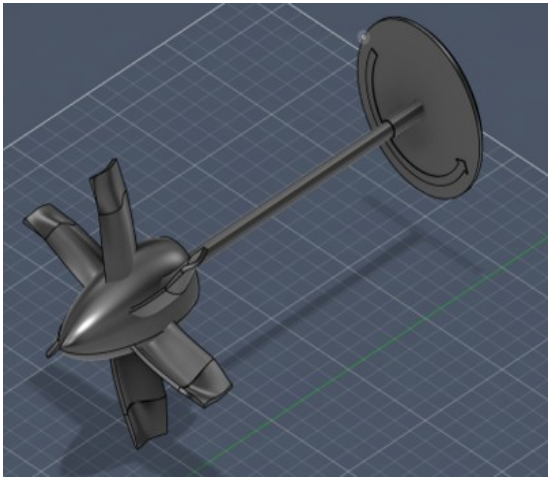


Fig. 1. Kaplan Turbine Without Blade Twist

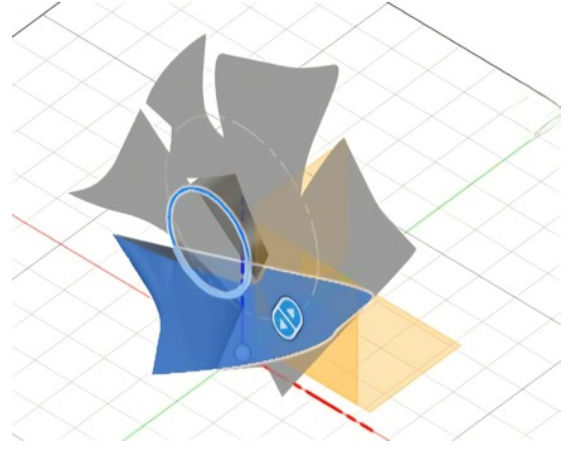


Fig. 2. Kaplan Turbine With Blade Twist

The twisted blade adapts to changing flow conditions, improving efficiency.

VI. CFD SIMULATION

The CFD simulation is carried out to analyze flow behavior inside the turbine.

A tetrahedral mesh is generated to discretize the flow domain.

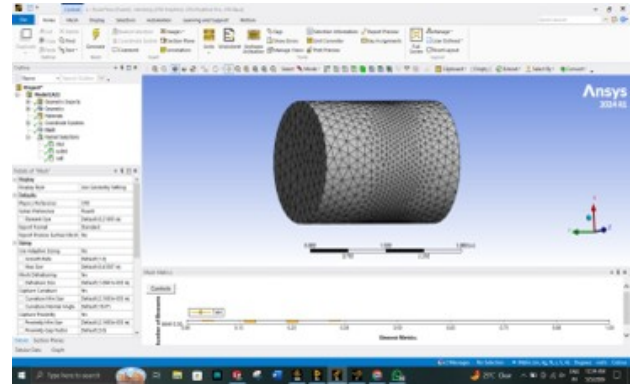


Fig. 3. Mesh Generated in ANSYS

A pressure-based solver with k-epsilon turbulence model is used.

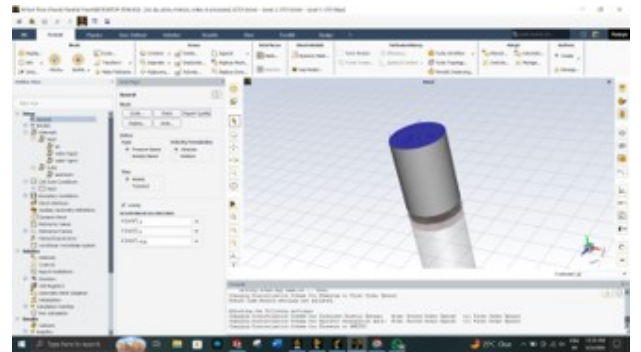


Fig. 4. Solver Setup

A. Simulation Parameters

TABLE II
CFD SIMULATION SETTINGS

Parameter	Value
Solver Type	Pressure-Based
Flow Type	Steady-State
Turbulence Model	k-epsilon
Mesh Type	Tetrahedral
Number of Nodes	426000
Velocity Formulation	Absolute
Working Fluid	Water

The domain includes inlet, runner, and outlet regions.

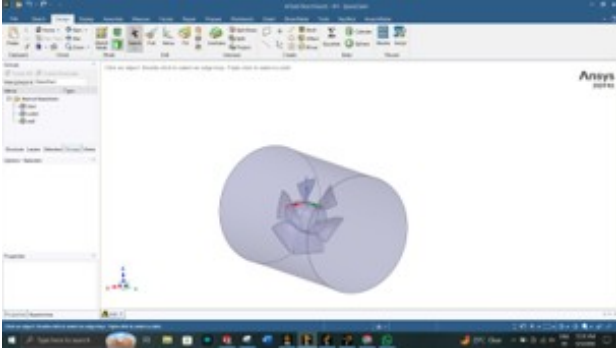


Fig. 5. Flow Domain

VII. RESULTS AND ANALYSIS

The performance of the Kaplan turbine is evaluated using CFD analysis by comparing two configurations: one without blade twist and one with blade twist. The comparison is carried out based on velocity distribution, pressure variation, and overall efficiency. The results provide a clear understanding of how blade geometry influences turbine performance.

A. Velocity Distribution

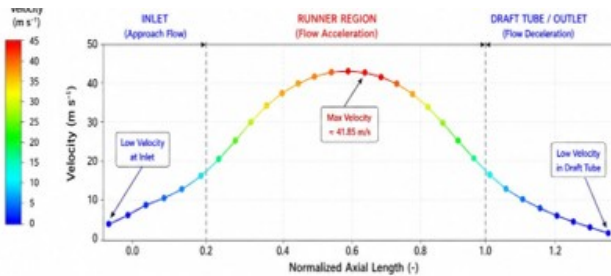


Fig. 6. Velocity Distribution Without Blade Twist

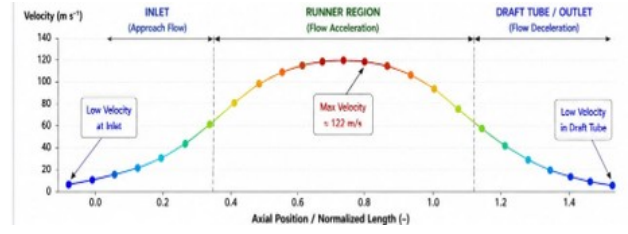


Fig. 7. Velocity Distribution With Blade Twist

The velocity distribution plots show a significant difference between the two configurations. In the turbine without blade twist, the velocity profile appears uneven and less concentrated, indicating inefficient interaction between the fluid and the blade surface. This results in lower energy transfer and reduced performance.

In contrast, the turbine with blade twist exhibits a much higher and more uniform velocity distribution. The fluid accelerates more effectively through the runner, indicating improved flow alignment with the blade surfaces. This enhancement in velocity directly contributes to better energy conversion and higher turbine efficiency.

B. Blade Geometry Parameters

TABLE III
BLADE GEOMETRY PARAMETERS

Parameter	Value
Runner Diameter	0.7 m
Hub Diameter	0.28 m
Blade Angle (Hub)	42°
Blade Angle (Tip)	28°
Chord Length (Hub)	0.12 m
Chord Length (Tip)	0.05 m

Additionally, the smoother velocity gradients observed in the twisted blade configuration suggest reduced turbulence and flow separation. This leads to minimized hydraulic losses and improved operational stability of the turbine.

C. Pressure Distribution

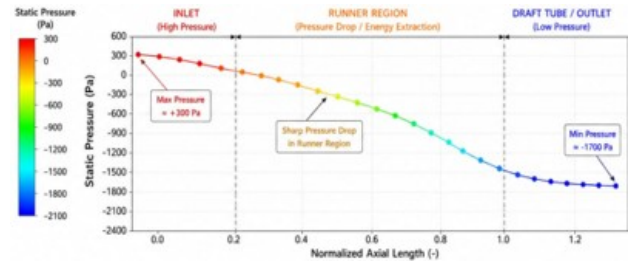


Fig. 8. Pressure Distribution Without Blade Twist

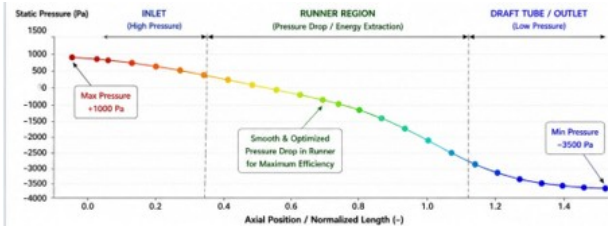


Fig. 9. Pressure Distribution With Blade Twist

The pressure distribution results further highlight the advantages of blade twist. In the untwisted blade configuration, the pressure variation across the turbine is relatively small, indicating that only a limited portion of the hydraulic energy is converted into mechanical work.

On the other hand, the twisted blade configuration shows a significantly larger pressure drop. This indicates that more pressure energy is being extracted from the fluid and converted into useful mechanical energy. The increased pressure difference across the blade surfaces confirms improved turbine performance.

It is also observed that both configurations exhibit regions of negative pressure, particularly near the blade surfaces. These regions are associated with flow separation and vortex formation, which may lead to cavitation in practical applications. However, the twisted blade design appears to manage these effects more effectively by maintaining better flow alignment.

D. Performance Comparison

TABLE IV
COMPARISON OF RESULTS

Parameter	Without Twist	With Twist
Max Velocity (m/s)	41.85	122
Min Velocity (m/s)	11.63	11.63
Max Pressure (Pa)	300	1000
Min Pressure (Pa)	-1700	-3500
Efficiency	0.7648	0.9338

The quantitative comparison clearly shows that the twisted blade turbine outperforms the untwisted configuration in all key parameters. The maximum velocity is significantly higher, indicating stronger energy transfer from fluid to the rotor.

Similarly, the larger pressure drop confirms that more hydraulic energy is being utilized effectively. Most importantly, the efficiency of the turbine increases from approximately 76% to 93%, which represents a substantial improvement.

These results strongly demonstrate that incorporating blade twist is essential for achieving high efficiency in Kaplan turbines. The improved flow alignment and reduced losses make the twisted blade configuration more suitable for real-world applications.

VIII. DISCUSSION

The results obtained from the CFD analysis clearly highlight the impact of blade twist on the performance of Kaplan

turbines. The velocity distribution plots indicate that the twisted blade configuration allows for smoother acceleration of fluid within the runner region. This improved velocity profile suggests more efficient energy transfer from the fluid to the rotating blades.

Similarly, the pressure distribution results demonstrate a significantly higher pressure drop in the twisted blade turbine. This is a key indicator of improved energy conversion, as a larger portion of the hydraulic energy is successfully transformed into mechanical work. In contrast, the untwisted blade configuration shows relatively smaller pressure gradients, indicating lower efficiency and higher energy losses.

Another important observation is the presence of negative pressure regions in both models. These regions are typically associated with flow separation and vortex formation, which may lead to cavitation in practical applications. However, the twisted blade design appears to manage these effects more effectively by maintaining better flow alignment and reducing turbulence intensity.

Overall, the analysis confirms that blade twist plays a crucial role in enhancing turbine performance. By adapting the blade orientation to match varying flow conditions along the radius, the twisted design minimizes losses and improves efficiency. These findings emphasize the importance of advanced blade design techniques in modern hydropower systems.

IX. NOVELTY OF THE WORK

This study combines computational design and CFD analysis into a unified framework. The comparison between twisted and untwisted blades under identical conditions provides clear insights.

X. CONCLUSION

This study presents a comprehensive investigation into the design and performance analysis of Kaplan turbine blades using a combination of computational methods and CFD simulation. Two turbine configurations, namely with and without blade twist, were developed and analyzed under identical operating conditions to evaluate the influence of blade geometry on overall performance.

The results obtained from the simulations clearly demonstrate that the twisted blade configuration outperforms the untwisted design in terms of velocity distribution, pressure drop, and efficiency. The improved alignment between the fluid flow and blade surface in the twisted design leads to more effective energy conversion and reduced hydraulic losses. This highlights the significance of incorporating blade twist in axial flow turbine design.

Furthermore, the study successfully establishes a link between theoretical design principles and practical CFD analysis. The integration of CAD modeling and numerical simulation provides a reliable framework for evaluating turbine performance before physical implementation. This approach not only reduces design time but also enhances accuracy.

In conclusion, blade geometry optimization, particularly through the incorporation of twist, plays a vital role in improving the efficiency of Kaplan turbines. The methodology

developed in this work can serve as a foundation for future research in turbine design, optimization, and performance enhancement.

XI. FUTURE WORK

Future work includes optimization, cavitation analysis, and experimental validation.

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